

LE CHUNG HIEU

DESIGN VERIFICATION INTERNSHIP

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PERSONAL STATEMENT

Final-year Electronics Engineering student specializing in IC Design and Verification. Experienced in SystemVerilog and UVM for developing coverage-driven verification environments, including constrained-random testing, functional coverage, and assertion-based verification. Hands-on experience with protocol verification (UART, APB) and hardware validation on FPGA platforms. Seeking a Verification Engineer internship to apply and further develop skills in UVM-based verification methodologies.

EDUCATION

Ho Chi Minh City University of Technology (VNU-HCMUT)	2022 – Current
Final-year student	GPA: 7.1/10
Faculty: Electrical and Electronics Engineering	Major: Electronics and Telecommunications Engineering

LANGUAGES

TOEIC (LISTENING/READING): 790/990

PROJECTS

PROTOCOL-LEVEL UART VERIFICATION WITH UVM AND SYSTEMVERILOG

Ho Chi Minh City University of Technology	Oct 2025 – Feb 2026
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Technologies: SystemVerilog, UVM, QuestaSim, Functional Coverage, Constrained-Random Verification, SystemVerilog Assertions.

Description: Developed a reusable UVM-based verification environment for UART IP, including key components such as agent, sequencer, driver, monitor, and a self-checking scoreboard. Implemented protocol-level verification supporting configurable baud rate, start/stop bits, and parity (even/odd). Designed a reference model within the scoreboard to ensure data integrity between DUT outputs and expected results. Applied coverage-driven verification methodology using both functional and cross coverage to validate multiple configuration scenarios. Created directed and constrained-random testcases, including error injection for parity and framing conditions. Integrated SystemVerilog Assertions (SVA) to enforce protocol compliance and detect functional violations. Delivered a scalable and reusable verification environment aligned with industry-standard UVM practices.

COVERAGE-DRIVEN VERIFICATION OF APB PROTOCOL USING UVM

Ho Chi Minh City University of Technology	Aug 2025 – Nov 2025
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Technologies: SystemVerilog, UVM, QuestaSim, AMBA APB Protocol, Functional & Cross Coverage, Constrained-Random Verification, SystemVerilog Assertions.

Description: Developed a UVM-based verification environment for the AMBA APB protocol, incorporating reusable components such as agent, driver, monitor, sequencer, and a self-checking scoreboard. Implemented protocol-aware driver and monitor to accurately handle APB transactions, including key signals like PSEL, PENABLE, PREADY, PWRITE, PRDATA, and PWDATA. Applied constrained-random verification techniques to generate diverse scenarios, covering back-to-back transfers and wait-state conditions. Built functional and cross coverage models to ensure thorough validation of address, data, and control signal combinations. Integrated SystemVerilog Assertions (SVA) to verify protocol compliance, particularly setup/enable phases and handshake timing. Developed multiple testcases, including basic read/write, random traffic, and stress tests, resulting in a scalable and reusable verification environment aligned with industry-standard UVM methodology.

CONSTRAINED-RANDOM VERIFICATION OF FIFO USING UVM

Ho Chi Minh City University of Technology

Oct 2025 – Jan 2026

Technologies: SystemVerilog, UVM, QuestaSim, Functional Coverage, Constrained-Random Verification, SystemVerilog Assertions, FIFO Design Verification.

Description: Built a UVM-based verification environment for a synchronous FIFO design, including driver, monitor, sequencer, and a self-checking scoreboard. Designed the scoreboard using a reference queue model to verify correct data ordering according to FIFO (First-In-First-Out) behavior. Applied constrained-random stimulus to validate various operational scenarios, including continuous write and read sequences. Verified critical corner cases such as overflow, underflow, and full/empty conditions. Developed a functional coverage model to ensure comprehensive exploration of read/write combinations and boundary states. Integrated SystemVerilog Assertions (SVA) to validate the correctness of status signals, including full, empty, and almost full. Created high-throughput stress tests to evaluate system robustness under extreme conditions, ensuring functional correctness across both normal and corner-case scenarios.

DESIGN AND VERIFICATION OF A 5-STAGE PIPELINED RISC-V PROCESSOR

Ho Chi Minh City University of Technology

Oct 2025 – Dec 2025

Technologies: SystemVerilog, Cadence Xcelium, Quartus, DE2-115 Altera FPGA Kit.

Description: Designed and implemented a 32-bit RV32I single-cycle RISC-V processor, including key modules such as PC, ALU, Register File, Control Unit, and memory. Extended the design to a 5-stage pipelined architecture with hazard detection and data forwarding mechanisms. Developed a SystemVerilog-based verification environment to validate functional correctness through simulation on Cadence Xcelium. Created directed and assembly-level test programs to verify instruction execution, pipeline behavior, and hazard handling. Performed waveform analysis and debugging to identify and resolve functional issues. Synthesized and deployed the design on a DE2-115 FPGA, validating hardware functionality by executing real applications such as a stopwatch.

SKILLS

- **Programming:** SystemVerilog, UVM, Verilog, VHDL.
- **Programming:** C, Python.
- **Scripting:** Shell Scripting.
- **Verification Skills:** Constrained-Random Verification, Functional & Cross Coverage, SystemVerilog Assertions (SVA), Scoreboarding.
- **Tools:** QuestaSim, ModelSim, Quartus, Vivado, Cadence Xcelium.
- **Others:** Linux, Git.